PRACTICAL FILE

IN

INFORMATICS PRACTICES-065

FOR

CLASS XI

SESSION 2026-27

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Certificate

This is to certify that _____ , Roll No. _____ of Class 11th Session 2026-27 has prepared the Practical File as per the prescribed Practical Syllabus of Informatics Practices Code-065, Class-11th under my supervision.

It is the original work done by him/her. His/her work is really appreciable. I wish him/her a very bright success

Signature of Teacher. _____

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Python Programs



Practical No-1

1. To find average and grade for given marks.

CODING:

```
sub1=int(input("Enter marks of the first subject: "))
sub2=int(input("Enter marks of the second subject: "))
sub3=int(input("Enter marks of the third subject: "))
sub4=int(input("Enter marks of the fourth subject: "))
sub5=int(input("Enter marks of the fifth subject: "))
tot=sub1+sub2+sub3+sub4+sub5
avg=(sub1+sub2+sub3+sub4+sub5)/5
print("Total",tot)
print("Average",avg)
if(avg>=90):
    print("Grade: A")
elif(avg>=80 and avg<90):
    print("Grade: B")
elif(avg>=70 and avg<80):
    print("Grade: C")
elif(avg>=60 and avg<70):
    print("Grade: D")
else:
    print("Grade: F")
```

OUTPUT:

Enter marks of the first subject: 80
Enter marks of the second subject: 90
Enter marks of the third subject: 95
Enter marks of the fourth subject: 96
Enter marks of the fifth subject: 97
Total 458
Average 91.6
Grade: A

Practical No-2

2A.To find sale price of an item with given cost and discount (%).

CODING:

```
cost = float(input("Enter the cost of an Item :- "))
disc = int(input("Enter the Discount (%) to be given :- "))
disc_amt = cost * disc/100
sale_price = cost-disc_amt
print("The cost price of an Item",cost)
print("Discount (%) given",disc)
print("Sale price is :- ",  sale_price)
```

OUTPUT

```
Enter the cost of an Item :- 100
Enter the Discount (%) to be given :- 20
The cost price of an Item 100.0
Discount (%) given 20
Sale price is :-  80.0
```

2B.To Swap numbers in python programme

CODING

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

print("Before swapping:")
print("a =", a)
print("b =", b)

temp = a
a = b
b = temp

print("After swapping:")
print("a =", a)
print("b =", b)
```

OUTPUT

```
Enter first number: 3
Enter second number: 4
Before swapping:
a = 3
b = 4
After swapping:
a = 4
b = 3
```

PROGRAMME-3

3.To calculate perimeter/circumference and area of shapes such as triangle, rectangle, square and circle.

CODING

```
print("Enter number to select shape:")
print("1.Triangle")
print("2.Rectangle")
print("3.Square")
print("4.Circle")
n = int(input("Enter your choice 1,2,3,4 :"))
if (n==1):
    S1 = int(input("Enter the length of Side 1 :"))
    S2 = int(input("Enter the length of Base :" ))
    S3 = int(input("Enter the length of Side 3 :"))
    h = int(input("Enter the height of the triangle : "))
    print ("Perimeter of trinagle is ",(S1+S2+S3))
    print ("Area of triangle is ",(1/2*S2*h))
elif (n==2):
    l = int(input("Enter the length of the rectangle : "))
    b = int(input("Enter the base of the rectangle : "))
    print("Perimeter of the rectangle is : ",(2*(l+b)))
    print("Area of the rectangle is : ",(l*b))
elif(n==3):
    s = int(input("Enter the length of the side of square : "))
    print("Perimeter of square is ",(4*s))
    print("Area of square is ",(s*s))
elif(n==4):
    r = int(input("Enter the radius of the circle : "))
    print("Circumference of the circle is ",(2*3.14*r))
    print("Area of the circle is ",(3.14*r*r))
else:
    print("Invalid Option")
```

OUTPUT

```
Enter number to select shape:
1.Triangle
2.Rectangle
3.Square
4.Circle
Enter your choice 1,2,3,4 :2
Enter the length of the rectangle : 22
Enter the base of the rectangle : 3
Perimeter of the rectangle is :  50
Area of the rectangle is :  66
```

PROGRAMME-4

4A. To calculate Simple and Compound interest.

CODING

```
principal = float(input('Enter amount: '))
rate = float(input('Enter rate: '))
time = float(input('Enter time: '))

# Calculation
simple_interest = (principal*time*rate)/100
compound_interest = principal * ( (1+rate/100)**time - 1)
# Displaying result
print('Simple interest is: %.2f' %(simple_interest))
print('Compound interest is: %.2f' %(compound_interest))
```

OUTPUT

```
Enter amount: 30
Enter rate: 1
Enter time: 5
Simple interest is: 1.50
Compound interest is: 1.53
```

4B.Write a python program to Display pyramid pattern

CODING

```
for i in range(1, 6):
    print(" " * (5 - i), end="")
    print("* " * i)
```

OUTPUT

```
    *
   * *
  * * *
 * * * *
* * * * *
```

4C.PYTHON PATTERN RIGHT ANGLED TRIANGLE

CODING

```
for i in range(1, 6):
    for j in range(1, i + 1):
        print("*", end=" ")
    print()
```

OUTPUT

```
*
* *
* * *
* * * *
* * * * *
```

PRACTICAL-5

5A.To calculate profit-loss for given Cost and Sell Price.

CODING

```
cp = float(input("Enter the Cost Price :- "))
sp = float(input("Enter the Selling Price :- "))
if sp > cp :
    profit = sp-cp
    profit_per = (profit * 100)//cp
    print("The Profit is :- ",profit)
    print("The Profit percentage  is :- ",profit_per)
elif sp <cp:
    loss = cp - sp
    loss_per = (loss * 100)//cp
    print("The Loss is :- ",loss)
    print("The Loss percentage is :- ",loss_per)
else:
    print("No Profit No Loss")
```

OUTPUT

```
Enter the Cost Price :- 35
Enter the Selling Price :- 40
The Profit is :-  5.0
The Profit percentage  is :-  14.0
```

5B.Write a program to check whether a number is positive, negative, or zero.

```
num = int(input("Enter a number: "))
if num > 0:
    print("The number is positive")
elif num < 0:
    print("The number is negative")
else:
    print("The number is zero")
```

OUTPUT

```
Enter a number: 0
The number is zero

Enter a number: 9
The number is positive

Enter a number: -8
The number is negative
```

PRACTICAL-6

6A. To calculate EMI for Amount, Period and Interest.

CODING

```
amount = float(input("Enter the Loan amount :- "))
rate = float(input("Enter  Interest Rate:- "))
period = float(input("Enter the time period :- "))
interest = amount * period * rate /100
total = amount + interest
emi = total /(period * 12)
print("EMI amount is ",emi)
```

OUTPUT

```
Enter the Loan amount :- 3000
Enter  Interest Rate:- 2
Enter the time period :- 3
EMI amount is  88.33333333333333
```

6B.Python programme using 1D,2D,3D and NDARRAY

CODING

```
import numpy as np
# 1D Array
a = np.array([10, 20, 30, 40])
print("1D Array:")
print(a)
print("Number of dimensions:", a.ndim)
# 2D Array
b = np.array([[10, 20, 30], [40, 50, 60]])
print("\n2D Array:")
print(b)
print("Number of dimensions:", b.ndim)
# 3D Array
c = np.array([[[1, 2], [3, 4]], [[5, 6], [7, 8]]])
print("\n3D Array:")
print(c)
print("Number of dimensions:", c.ndim)
```

OUTPUT

```
1D Array: [10 20 30 40]
Number of dimensions: 1
2D Array:
[[10 20 30]
 [40 50 60]]
Number of dimensions: 2
3D Array:
[[[1 2]
  [3 4]]
 [[5 6]
  [7 8]]]
Number of dimensions: 3
```

PRACTICAL-7

7A. To calculate tax - GST / Income Tax.

CODING

```
cost = float(input("Enter cost of the product/service:- "))
gstper = float(input("Enter the % charged for gst :- "))
gst = cost * gstper/100
amount = cost + gst
print("GST is :- ",gst)
print("Amount to Pay is :- ",amount)
# Income Tax
income = float(input("Enter total annual income :-  "))
if income <20000:
    tax = 0
elif income <400000:
    tax = (income-200000)*0.05
elif income <800000:
    tax = (income-400000)*0.07 + 10000
else :
    tax = (income-800000)*0.1 +38000
print("Income Tax to pay",tax)
```

OUTPUT

```
Enter cost of the product/service:- 100
Enter the % charged for gst :- 3
GST is :-  3.0
Amount to Pay is :-  103.0
Enter total annual income :-  130000
Income Tax to pay -3500.0
```

PRACTICAL-8

8A. To find the largest and smallest numbers in a list.

CODING

```
lst = []
num = int(input('How many numbers: '))
for n in range(num):
    numbers = int(input('Enter number '))
    lst.append(numbers)
print("Maximum(Largest) element in the list is :", max(lst))
print("\nMinimum(Smallest) element in the list is :", min(lst))
```

OUTPUT

```
How many numbers: 5
Enter number 2
Enter number 4
Enter number 5
Enter number 67
Enter number 88
Maximum(Largest) element in the list is : 88
Minimum(Smallest) element in the list is : 2
```

8B. To find the third largest/smallest number in a list

CODING

```
def find_len(list1):
    length = len(list1)
    list1.sort()
    print("Given list is", list1)
    print("Largest element is:", list1[length-1])
    print("Smallest element is:", list1[0])
    print("Third Smallest element is:", list1[2])
    print("Third Largest element is:", list1[length-3])
# LIST
list1=[12, 45, 2, 41, 31, 10, 8, 6, 4]
Largest = find_len(list1)
```

OUTPUT

```
Given list is [2, 4, 6, 8, 10, 12, 31, 41, 45]
Largest element is: 45
Smallest element is: 2
Third Smallest element is: 6
Third Largest element is: 31
```

PRACTICAL-9

9.WRITE A PYTHON PROGRAMME USING OPERATOR PRECEDENCE

```
# Operator Precedence in Python
print("1. Parentheses")
print("(10 + 5) * 2 =", (10 + 5) * 2)

print("\n2. Exponentiation")
print("2 ** 3 =", 2 ** 3)

print("\n3. Multiplication, Division, Floor Division, Modulus")
print("10 * 2 =", 10 * 2)
print("10 / 2 =", 10 / 2)
print("10 // 3 =", 10 // 3)
print("10 % 3 =", 10 % 3)

print("\n4. Addition and Subtraction")
print("10 + 5 - 2 =", 10 + 5 - 2)

print("\n5. Relational Operators")
print("10 > 5 =", 10 > 5)
print("10 == 5 =", 10 == 5)

print("\n6. Logical Operators")
print("10 > 5 and 8 > 3 =", 10 > 5 and 8 > 3)
print("10 < 5 or 8 > 3 =", 10 < 5 or 8 > 3)
print("not 10 > 5 =", not 10 > 5)

print("\n7. Bitwise Operators")
print("5 & 3 =", 5 & 3)
print("5 | 3 =", 5 | 3)
print("5 ^ 3 =", 5 ^ 3)

print("\n8. Assignment Operators")
a = 10
print("Initial value of a =", a)
a += 5
print("After a += 5, a =", a)
a *= 2
print("After a *= 2, a =", a)
```

OUTPUT

1. Parentheses
(10 + 5) * 2 = 30

2. Exponentiation
2 ** 3 = 8

3. Multiplication, Division, Floor Division, Modulus
10 * 2 = 20
10 / 2 = 5.0
10 // 3 = 3
10 % 3 = 1

4. Addition and Subtraction
10 + 5 - 2 = 13

5. Relational Operators
10 > 5 = True
10 == 5 = False

6. Logical Operators
10 > 5 and 8 > 3 = True
10 < 5 or 8 > 3 = True
not 10 > 5 = False

7. Bitwise Operators
5 & 3 = 1
5 | 3 = 7
5 ^ 3 = 6

8. Assignment Operators
Initial value of a = 10
After a += 5, a = 15
After a *= 2, a = 30

PRACTICAL-10

10.PYTHON PROGRAMME USING DICTIONARY MANIPULATION CODING

```
student = {"Name": "Rafiq", "Class": 11, "Marks": 85}
print("Original Dictionary:", student)

# 1. Add a new item
student["Section"] = "A"
print("After adding an item:", student)

# 2. Modify an existing value
student["Marks"] = 95
print("After modifying marks:", student)

# 3. Delete an item using del
del student["Section"]
print("After deleting an item:", student)

# 4. Delete an item using pop()
student.pop("Class")
print("After using pop():", student)

# 5. Display keys
print("Keys:", student.keys())

# 6. Display values
print("Values:", student.values())

# 7. Display key-value pairs
print("Items:", student.items())


# 8. Search for a key
key = input("Enter key to search: ")
if key in student:
    print("Key exists")
    print("Value:", student[key])
else:
    print("Key does not exist")

# 9. Count dictionary items
print("Number of items:", len(student))
```

```
# 10. Traverse the dictionary
print("Dictionary items:")
for key, value in student.items():
    print(key, ":", value)

# 11. Clear the dictionary
student.clear()
print("After clearing:", student)
```

OUTPUT

```
Original Dictionary: {'Name': 'Rafiq', 'Class': 11, 'Marks': 85}

After adding an item: {'Name': 'Rafiq', 'Class': 11, 'Marks': 85,
'Section': 'A'}

After modifying marks: {'Name': 'Rafiq', 'Class': 11, 'Marks': 95,
'Section': 'A'}

After deleting an item: {'Name': 'Rafiq', 'Class': 11, 'Marks': 95}

After using pop(): {'Name': 'Rafiq', 'Marks': 95}

Keys: dict_keys(['Name', 'Marks'])

Values: dict_values(['Rafiq', 95])

Items: dict_items([('Name', 'Rafiq'), ('Marks', 95)])
```

PRACTICAL-11

11. (A)To print the first 'n' multiples of given number using FOR loop

CODING

```
num = int(input("Enter a number: "))
for i in range(1, 11):
    print(num, "x", i, "=", num * i)
```

(B)To print the first 'n' multiples of given number using WHILE loop

```
num = int(input("Enter a number: "))
i = 1
while i <= 10:
    print(num, "x", i, "=", num * i)
    i = i + 1
```

OUTPUT

Enter a number: 12 (Write 12 th table as output)

C. Python Program using for and while loops

```
n = int(input("Enter a number: "))
# 1. Display numbers using for loop
print("\nNumbers using for loop:")
for i in range(1, n + 1):
    print(i, end=" ")
# 2. Display numbers using while loop
print("\n\nNumbers using while loop:")
i = 1
while i <= n:
    print(i, end=" ")
    i = i + 1
# 3. Sum of numbers using for loop
total = 0
for i in range(1, n + 1):
    total = total + i
print("\n\nSum using for loop:", total)
# 4. Sum of numbers using while loop
total = 0
i = 1
while i <= n:
    total = total + i
    i = i + 1
print("Sum using while loop:", total)
```

OUTPUT

```
Enter a number: 1
Numbers using for loop:1
Numbers using while loop:1
Sum using for loop: 1
Sum using while loop: 1
```

PRACTICAL-12

12. To count the number of vowels in user entered string

CODING

```
s=input("Enter Sentence")
str=s.lower()
count=0
list=['a','e','i','o','u']
for char in str:
    if char in list:
        count=count+1
print("no.of vowels",count)
```

OUTPUT

```
Enter SentenceSYED RAFIQ
no.of vowels 3
```

12B. Count and display the number of vowels, consonants, uppercase, lowercase ,characters in string.

CODING

```
s = input("Enter a string: ")
v = 0
c = 0
u = 0
l = 0
for ch in s:
    if ch in "AEIOUaeiou":
        v = v + 1
    elif ch.isalpha():
        c = c + 1

    if ch.isupper():
        u = u + 1
    elif ch.islower():
        l = l + 1

print("Number of vowels:", v)
print("Number of consonants:", c)
print("Number of uppercase characters:", u)
print("Number of lowercase characters:", l)
```

OUTPUT

```
Enter a string: SYed RAFIQ
Number of vowels: 3
Number of consonants: 6
Number of uppercase characters: 7
Number of lowercase characters: 2
```

PRACTICAL-13

13.WRITE A PYTHON PROGRAMME USING 5 OPERATORS.

CODING

```
a = 10
b = 3
print("A =", a);print("B =", b)
# 1. Arithmetic Operators
print("\nArithmetic Operators");
print("Addition:", a + b)                    print("Subtraction:", a - b);
print("Multiplication:", a * b)  ;           print("Division:", a / b)
print("Floor Division:", a // b);            print("Modulus:", a % b)
print("Exponentiation:", a ** b)
# 2. Relational Operators
print("\nRelational Operators")
print("a > b:", a > b) ;                      print("a < b:", a < b)
print("a == b:", a == b);                    print("a != b:", a != b)
print("a >= b:", a >= b)                     print("a <= b:", a <= b)
# 3. Logical Operators
print("\nLogical Operators")
print("a > 5 and b < 5:", a > 5 and b < 5)
print("a < 5 or b < 5:", a < 5 or b < 5)
print("not(a > b):", not(a > b))
# 4. Assignment Operators
print("\nAssignment Operators")
x = 10
x += 5
print("After x += 5:", x)
x -= 3
print("After x -= 3:", x)
x *= 2
print("After x *= 2:", x)
x //= 4
print("After x //= 4:", x)
# 5. Membership Operators
print("\nMembership Operators")
name = "RAFIQ"
print("'R' in name:", "R" in name)
print("'Z' not in name:", "Z" not in name)
```

OUTPUT

```
A = 10
B = 3

Arithmetic Operators
Addition: 13
Subtraction: 7
Multiplication: 30
Division: 3.3333333333333335
Floor Division: 3
Modulus: 1
Exponentiation: 1000

Relational Operators
a > b: True
a < b: False
a == b: False
a != b: True
a >= b: True
a <= b: False

Logical Operators
a > 5 and b < 5: True
a < 5 or b < 5: True
not(a > b): False

Assignment Operators
After x += 5: 15
After x -= 3: 12
After x *= 2: 24
After x //= 4: 6

Membership Operators
'R' in name: True
'Z' not in name: True
```

PRACTICAL-14

14. PYTHON PROGRAMME USING MUTABLE AND IMMUTABLE DATA TYPE.

CODING

```
# Mutable and Immutable Data Types
# 1. Mutable Data Type - List
print("MUTABLE DATA TYPE - LIST")
L = [10, 20, 30]
print("Original List:", L)
L[0] = 100
print("After modification:", L)
L.append(40)
print("After adding an element:", L)
print()

# 2. Mutable Data Type - Set
print("MUTABLE DATA TYPE - SET")
S = {10, 20, 30}
print("Original Set:", S)
S.add(40)
print("After adding an element:", S)
S.remove(20)
print("After removing an element:", S)
print()

# 3. Immutable Data Type - String
print("IMMUTABLE DATA TYPE - STRING")
s = "Python"
print("Original String:", s)
# Strings cannot be modified character by character
s = s + " Programming"
print("After reassignment:", s)
print()

# 4. Immutable Data Type - Tuple
print("IMMUTABLE DATA TYPE - TUPLE")
T = (10, 20, 30)
print("Original Tuple:", T)
# Tuples cannot be modified
# T[0] = 100  # This causes TypeError
T = T + (40,)
print("After reassignment:", T)
print()
# 5. Immutable Data Type - Integer
print("IMMUTABLE DATA TYPE - INTEGER")
a = 10
print("Original value:", a)
a = a + 5
print("After reassignment:", a)
```

OUTPUT

```
MUTABLE DATA TYPE - LIST
Original List: [10, 20, 30]
After modification: [100, 20, 30]
After adding an element: [100, 20, 30, 40]

MUTABLE DATA TYPE - SET
Original Set: {10, 20, 30}
After adding an element: {40, 10, 20, 30}
After removing an element: {40, 10, 30}

IMMUTABLE DATA TYPE - STRING
Original String: Python
After reassignment: Python Programming

IMMUTABLE DATA TYPE - TUPLE
Original Tuple: (10, 20, 30)
After reassignment: (10, 20, 30, 40)

IMMUTABLE DATA TYPE - INTEGER
Original value: 10
After reassignment: 15
```

PRACTICAL-15

15.Create a dictionary to store names of states and their capitals.

CODING

```
#Create a dictionary to store names of states and their capitals
States = dict()
n = int(input("Enter number of states : "))
for i in range(n):
   State = input("Enter State : ")
   Capital = input("Enter Capital : ")
   States[State] = Capital
print("States Dictionary : ",States)
```

OUTPUT

```
Enter number of states : 5
Enter State : AP,Enter Capital : AM
Enter State : TG,Enter Capital : HY
Enter State : KA,Enter Capital : BL
Enter State : TN,Enter Capital : CH
Enter State : MH,Enter Capital : BO
States Dictionary :  {'AP': 'AM', 'TG': 'HY', 'KA': 'BL', 'TN': 'CH', 'MH': 'BO'}
```

15B. Create a dictionary of students to store names and marks obtained in 5 subjects.

CODING

```
students = {
    "Ravi": [85, 90, 78, 88, 92],
    "Sita": [92, 95, 89, 91, 94],
    "Aman": [76, 80, 72, 85, 79],
    "Priya": [88, 86, 90, 84, 93]
}

print("Student Details")
for name, marks in students.items():
    print("Name:", name)
    print("Marks:", marks)
    print("Total:", sum(marks))
    print("Average:", sum(marks) / 5)
    print("--------------------")
```

OUTPUT

```
Student Details
Name: Ravi,Marks: [85, 90, 78, 88, 92],Total: 433,Average: 86.6
Name: Sita,Marks: [92, 95, 89, 91, 94],Total: 461,Average: 92.2
Name: Aman,Marks: [76, 80, 72, 85, 79],Total: 392,Average: 78.4
Name: Priya,Marks: [88, 86, 90, 84, 93],Total: 441,Average: 88.2
--------------------
```

PRACTICAL-16

16. To print the highest and lowest values in the dictionary.

CODING

```
s={"Rafiq":50,"Syed":20,"Chand":30,"Asha":10,"Affan":40}
high=max(s,key=s.get)
hmarks=max(s.values())
print("Student with highest marks is ",high)
low=min(s,key=s.get)
hmarks=min(s.values())
print("Students with lowest marks  is",low)
```

OUTPUT

```
Student with highest marks is  Rafiq
Students with lowest marks  is Asha
```

16B.Create a dictionary with the roll number, name and marks of n students in a class and display the names of students who have marks above 75.

CODING

```
n = int(input("Enter number of students: "))
students = {}
for i in range(n):
    rno = int(input("Enter roll number: "))
    name = input("Enter student name: ")
    marks = int(input("Enter marks: "))
    students[rno] = [name, marks]

print("Students who scored above 75:")

for rno in students:
    if students[rno][1] > 75:
        print(students[rno][0])
```

OUTPUT

```
Enter number of students: 3
Enter roll number: 1
Enter student name: A
Enter marks: 78
Enter roll number: 2
Enter student name: B
Enter marks: 99
Enter roll number: 3
Enter student name: C
Enter marks: 23
Students who scored above 75:
A
B
```